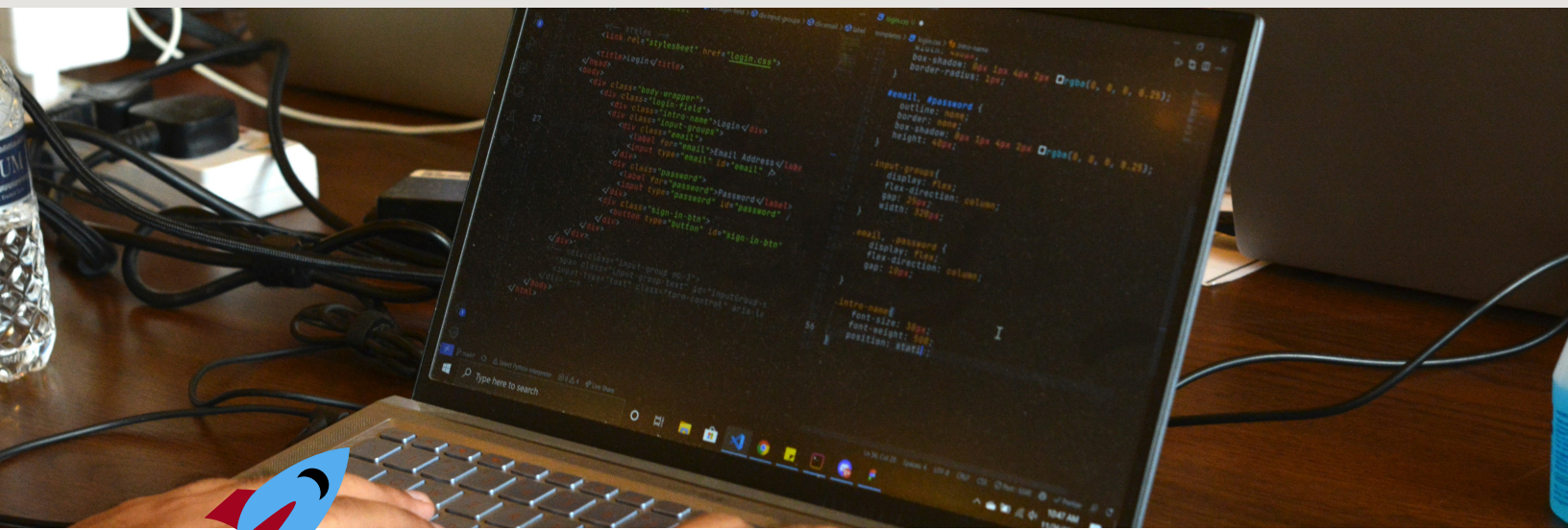


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hoichoi Hackathon'26 — Participant Handbook

📌 **Build production-grade AI systems for real problems in streaming, content, and media operations.**

📅 **Date:** 26th September 2026 · ⌚ **Time:** 11:30 am IST · 📍 **Location:** Virtual · Solo participation only

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About the Hackathon

Welcome to the hoichoi Hackathon — a solo-participation virtual event bringing together students, developers, and AI/ML builders to solve real problems from streaming, content ops, and media pipelines, and build production-grade AI systems from scratch.

This is not a team event. Each participant builds and submits solo, working through **one** problem statement of their choice, end to end.

What to expect:

- Real-world problem statements drawn from streaming, content ops, and media pipelines
- What "production-grade" actually means — and what judges look for
- Tips for getting started — tools, approach, and scoping your solve
- A live, working demo link and GitHub repo required at submission
- Placement and hiring opportunities for standout participants — that's the real prize here

Timeline

Milestone	Date / Time
Hackathon starts	26th September 2026, 11:30 am IST
Duration	12 hours
Submission window closes	12:30 am IST (1 hour after the 12-hour mark)
Results announced	Within 2 weeks of the hackathon

Rules

- **Solo participation only** — no teams. Each participant builds and submits individually.
- **Pick one problem statement** — choose any one of the four below and solve it end to end. You do not need to attempt more than one.
- **Judging criteria:**
 - **AI-native approach** — how effectively you use AI as the core of your solution, not a bolt-on feature
 - **Efficient, working solve** — a real, functioning answer to the problem statement, built end to end

Problem Statements

Choose **one** of the following four to solve.

▼ Problem 1 — Context-Aware Video Segmentation & Intelligent Ad Placement

Track: Video Understanding / AdTech

Ingest a long-form Bengali drama, segment it into semantically coherent scenes, score which scene boundaries are safe to interrupt, and match each surviving break to the most contextually appropriate brand from a synthetic catalogue — then emit a standards-compliant ad-break manifest and a playable demo.

Must get right

Where	Is this timestamp a natural, non-jarring place to cut? (mid-sentence cuts are heavily penalised)
Whether	Is a break warranted here at all, given pacing rules (max breaks/hour, min gap, ad load %)?
What	Which brand's creative belongs in this specific slot — dominant scene activity wins, negative_contexts are a hard block, not a soft penalty

MVP scope

Video in → scenes segmented → break candidates scored → brand matched (must generalise to a 9th, unseen brand with zero code changes) → VMAP manifest + debug JSON + playable demo that actually cuts to the ad and resumes.

Toughest test

Held-out videos are chosen specifically to punish naive scoring: judges check for mid-dialogue cuts and blind-viewing scene quality, and any negative-context brand placed against a blocked scene (e.g. a food ad after a funeral) in the held-out set scores that criterion **zero**, not low.

Auto-disqualifiers

- Hard-coding timestamps or brand assignments for the sample videos
- Any negative-context violation in the held-out set
- Demo that cannot run live at presentation time
- Substituting real company names for the synthetic brands

Resources: [Content & sample assets folder](#)

▼ Problem 2 — Automated Bengali Subtitle & Closed-Caption Pipeline with Speaker Diarization

Track: Speech & Language / Media Localisation

Build an end-to-end pipeline that turns a Bengali-language video into a broadcast-quality Bengali closed-caption track (speaker-attributed, with non-speech sound events) plus translated English and Hindi subtitle tracks — timed to industry timed-text standards, with a QC report that tells a human exactly which cues to review.

Must get right

Recognition	What words were actually said, in Bengali — including correctly handling code-switched English inline (e.g. "ওর office-এ একটা meeting আছে")
Attribution	Who said it, and exactly when — speaker IDs must stay stable across the whole runtime, no identity swaps
Presentation	Can a human actually read this on screen — CPS limits, line length, no straddling a shot change

MVP scope

Video in → Bengali ASR + diarization → speaker naming (cast list / vocative cues - optional) → forced alignment → cue segmentation → Bengali CC (WebVTT) + EN/HI subtitles (SRT) + QC report with a ranked review queue.

Toughest test

Held-out episodes are checked directly for hallucinated subtitle text over silence or music, unflagged — this is the failure mode that makes automated captioning unshippable. The QC report's review queue is also compared against the actual error list, not just accepted at face value.

Auto-disqualifiers

- Shipping a hand-corrected transcript for the sample episodes
- Unflagged hallucinated text over silence/music in the held-out set
- Demo that cannot run live at presentation time

Resources: [Content & sample assets folder](#)

▼ Problem 3 — AI Content Studio & Multi-Platform Command Center

Track: Content Ops / Generative AI

Four integrated modules — Generative Studio, Publisher, Analytics Store, Cross-Platform Insights — that turn a single content brief into on-brand generated assets, platform-tailored captions, scheduled posts, unified engagement metrics, and a weekly report that feeds insights back into the next brief.

Must get right

Generation	Visual and copy must differ meaningfully per channel (tone, length, hashtag convention, CTA) — Bengali and English are both natively generated, not translated
Approval gate	Nothing enters the publishing queue without explicit human approval; a discard-and-retry loop exists before that point
Comparison	Cross-platform insights compare like-for-like content side by side, not just separate per-platform totals

MVP scope

Brief in → generated image/video + tailored copy for 3 channels → approved → scheduled → "published" via mock channel adapter → metrics ingested → one cross-platform comparison + one weekly AI-written report, every claim citing post IDs.

Toughest test

A Bengali brief is checked for genuinely native-sounding output, not machine translation. A post that violates a platform constraint (aspect ratio, file size, character limit) is submitted to the adapter layer and must be rejected, not silently accepted.

Auto-disqualifiers

- One generated image cropped/relabelled per platform and called "tailored"
- Generation with no approval gate before scheduling
- Insights that live on a dashboard but never reach brief creation

▼ Problem 4 — Creative Reformatting Engine

Track: Media Pipeline / Computer Vision

A media pipeline that takes one master image and/or video for a title and produces every platform-ready ratio (16:9, 1:1, 9:16, 4:5) with subject-aware smart crop, a still pulled cleanly from video, and a subject-tracked, active-speaker-aware vertical reel — every derived asset validated against a machine-readable platform spec before it enters the library.

Must get right

Crop	Subject-aware smart crop that detects faces/subjects and composes around them — not the frame's centre
Reframe	Video reframing tracks the subject as they move; when multiple people are on screen, the crop follows whoever is speaking, in sync with audio
Validate	Every generated asset gets an automated pass/fail compliance report against the spec sheet before entering the library — no asset enters unvalidated

MVP scope

One master image → 4 subject-aware ratio variants → validated → in the library, with single-variant regenerate working. One master video → one speaker-aware reel-format cutdown, plus one still extracted as an image variant.

Toughest test

Test assets are chosen specifically to punish naive cropping (off-centre subjects, multiple faces) and every variant is inspected for cropped faces and violated safe zones. A two-person alternating-speaker clip is reframed and checked that the crop follows whoever is talking, not a fixed wide frame. Judges validate outputs against the spec sheet themselves and compare against the system's own verdict.

Auto-disqualifiers

- Center-crop with a face-detection call bolted on that doesn't actually influence the crop box
- Vertical video reframed once at frame 0 and held static for the whole clip
- A validator that only checks dimensions

How to Submit

Resources: [Content & sample assets folder](#)

Your submission must include all three of the following:

1. **A live, working demo link** — something judges can actually open and test
2. **A public GitHub repository link** — your code, viewable by judges
3. **A brief explanatory video (under 5 minutes)** — walking through your